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Experiment No-12

AIM: Retrieve and Display Data from Public API in

Flutter

Program :

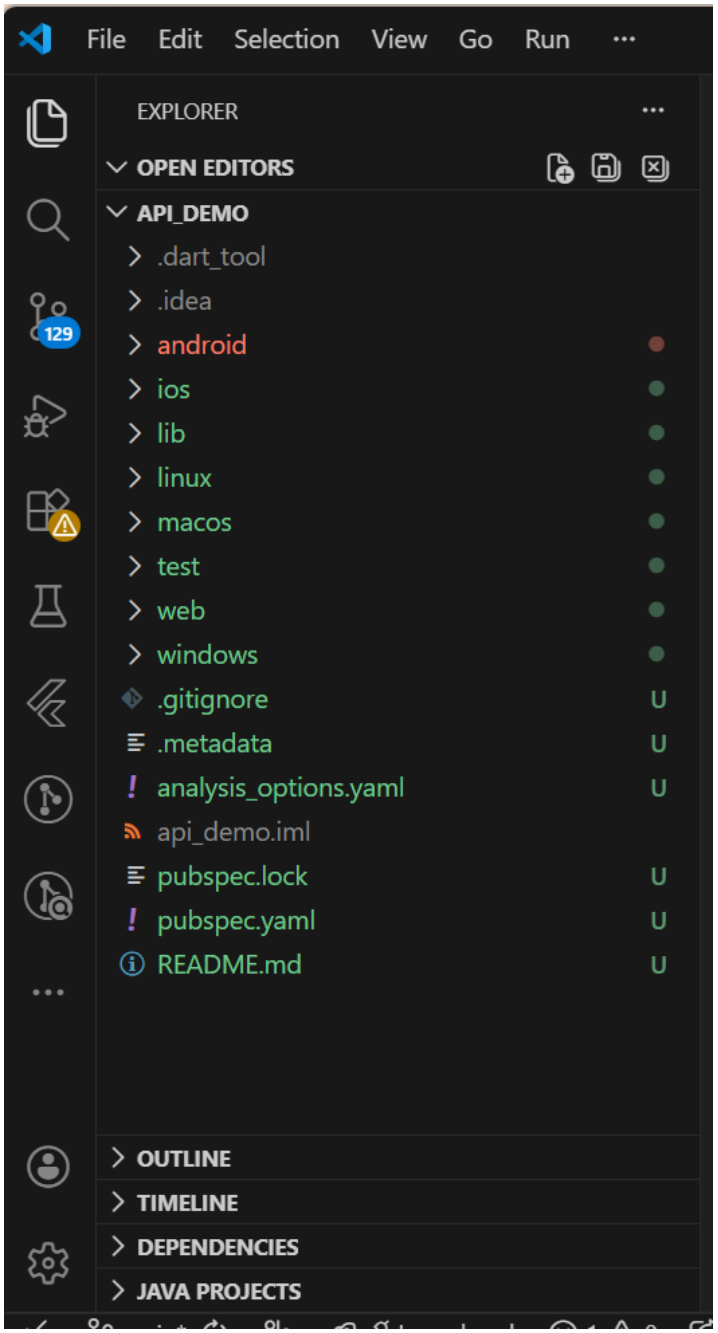
Open terminal:

flutter create api_demo

cd api_demo

code .

```
PS C:\ecommerce1_pwa\ecommerce2_pwa> flutter create api_demo
>> cd api_demo
Creating project api_demo...
Resolving dependencies in `api_demo`... (1.0s)
Downloading packages...
Got dependencies in `api_demo`.
Wrote 130 files.

All done!
You can find general documentation for Flutter at: https://docs.flutter.dev/
Detailed API documentation is available at: https://api.flutter.dev/
If you prefer video documentation, consider: https://www.youtube.com/c/flutterdev

In order to run your application, type:

  $ cd api_demo
  $ flutter run

Your application code is in api_demo\lib\main.dart.

PS C:\ecommerce1_pwa\ecommerce2_pwa\api_demo> code .
PS C:\ecommerce1_pwa\ecommerce2_pwa\api_demo>
```



Step 3: Add HTTP Package

Open pubspec.yaml and add:

dependencies:

  flutter:

    sdk: flutter

http: ^0.13.6

Then run:

flutter pub get

```
PS C:\ecommerce1_pwa\ecommerce2_pwa\api_demo> flutter pub get
Resolving dependencies...
Downloading packages...
  characters 1.4.0 (1.4.1 available)
  http 0.13.6 (1.6.0 available)
  matcher 0.12.17 (0.12.19 available)
  material_color_utilities 0.11.1 (0.13.0 available)
  meta 1.17.0 (1.18.2 available)
  test_api 0.7.7 (0.7.11 available)
  vector_math 2.2.0 (2.3.0 available)
Got dependencies!
7 packages have newer versions incompatible with dependency constraints.
Try `flutter pub outdated` for more information.
PS C:\ecommerce1_pwa\ecommerce2_pwa\api_demo>
```

Step 4: Choose Public API

Example API:

https://jsonplaceholder.typicode.com/posts

jsonplaceholder.typicode.com/posts

Pretty-print


```
[
  {
    "userId": 1,
    "id": 1,
    "title": "sunt aut facere repellat provident occaecati excepturi optio reprehenderit",
    "body": "quia et suscipit\nsuscipit recusandae consequuntur expedita et cum\nreprehenderit molestiae ut ut quas totam\nnostrum rerum est autem sunt rem eveniet architecto"
  },
  {
    "userId": 1,
    "id": 2,
    "title": "qui est esse",
    "body": "est rerum tempore vitae\nsequi sint nihil reprehenderit dolor beatae ea dolores neque\nfugiat blanditiis voluptate porro vel nihil molestiae ut reiciendis\nqui
aperiam non debitis possimus qui neque nisi nulla"
  },
  {
    "userId": 1,
    "id": 3,
    "title": "ea molestias quasi exercitationem repellat qui ipsa sit aut",
    "body": "et iusto sed quo iure\nvoluptatem occaecati omnis eligendi aut ad\nvoluptatem doloribus vel accusantium quis pariatur\nmolestiae porro eius odio et labore et velit
aut"
  },
  {
    "userId": 1,
    "id": 4,
    "title": "eum et est occaecati",
    "body": "ullam et saepe reiciendis voluptatem adipisci\nsit amet autem assumenda provident rerum culpa\nquis hic commodi nesciunt rem tenetur doloremque ipsam iure\nquis
sunt voluptatem rerum illo velit"
  },
  {
    "userId": 1,
    "id": 5,
    "title": "nesciunt quas odio",
    "body": "repudiandae veniam quaerat sunt sed\nalias aut fugiat sit autem sed est\nvoluptatem omnis possimus esse voluptatibus quis\nest aut tenetur dolor neque"
  },
  {
```

Step 6: Write Code to Fetch Data

Open main.dart and write:

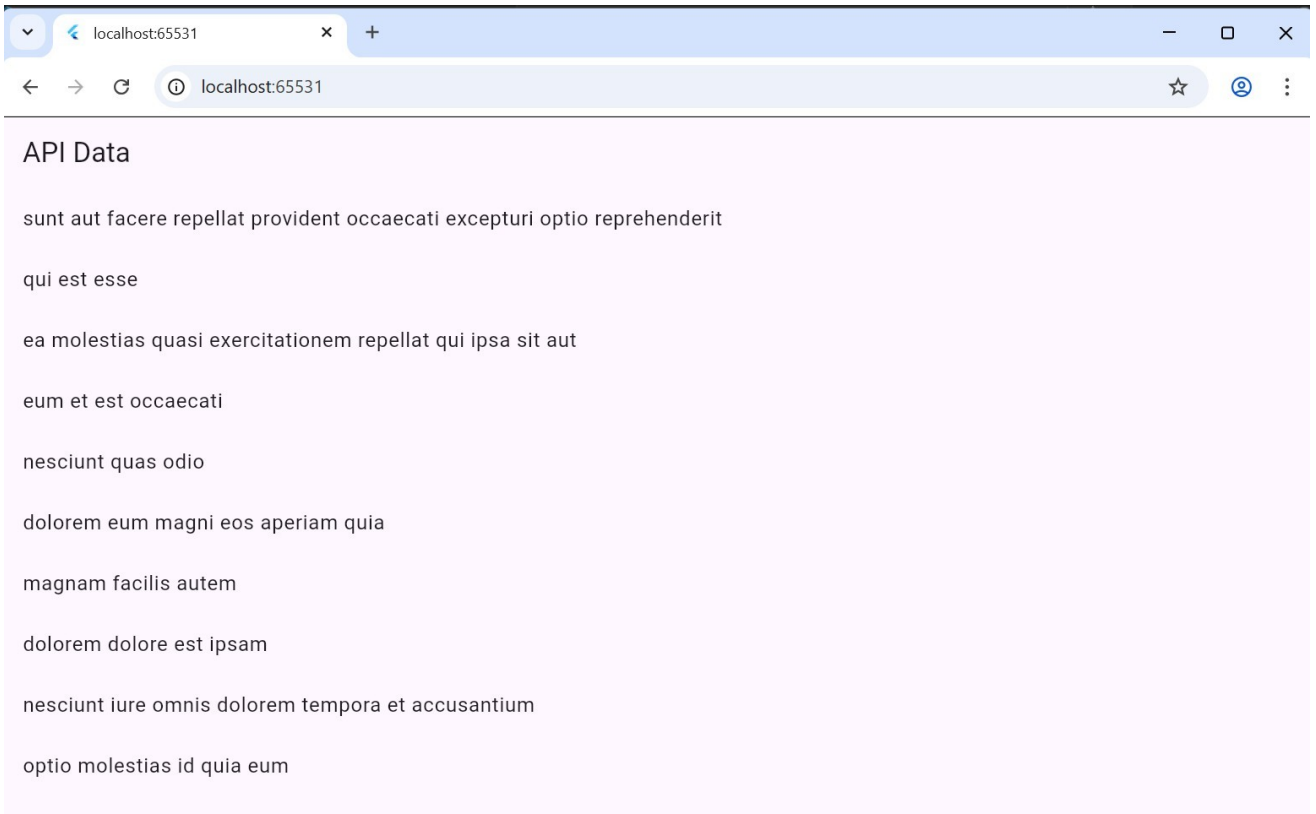
```
import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;
import 'dart:convert';

void main() {
  runApp(MaterialApp(home: MyApp()));
}

class MyApp extends StatelessWidget {

  // Function to fetch data
  Future<List> fetchData() async
  { final response = await http.get(
      Uri.parse('https://jsonplaceholder.typicode.com/posts'),
    );

    if (response.statusCode == 200)
      { return json.decode(response.body);
    } else {
      throw Exception("Failed to load data");
    }
  }

  @override
  Widget build(BuildContext context)
    { return Scaffold(
      appBar: AppBar(title: Text("Simple API App")),
```

```
body: FutureBuilder(
  future: fetchData(),
  builder: (context, snapshot) {

    // Loading
    if (snapshot.connectionState == ConnectionState.waiting)
      { return Center(child: CircularProgressIndicator());
    }

    // Error
    if (snapshot.hasError) {
      return Center(child: Text("Error: ${snapshot.error}"));
    }

    // Data
    final data = snapshot.data as List;

    return
      ListView.builder( itemCount:
      data.length, itemBuilder:
      (context, index) { return
      ListTile(
          title: Text(data[index]['title']),
          subtitle: Text(data[index]['body']),
        );
      },
    );
  },
```

),

Conclusion – In this way we have implemented the app,which will Fetch real-time data from an API and it will Parse JSON into Dart objects and also Dynamically update UI with API data.

```
  );
 }
}
```

Step 7: Run the Application

flutter run